

1. 设 $g(x) = (f'(x))^2 - 2Lf(x)$ $\therefore g'(x) = 2f'(x)f''(x) - 2Lf'(x) = 2f'(x)(f''(x) - L)$

又: $f(x) > 0, x \in \mathbb{R}$ $\therefore$ 下分两种情况

① $f(x)$ 存在最小值, 设 $f(x)_{\min} = f(x_0)$ $\therefore f(x) -$

1. 设 $g(x) = \frac{(f'(x))^2}{f(x)}$ $\therefore g'(x) = \frac{f'(x)[2f'(x)f''(x) - (f'(x))^2]}{f^2(x)}$

对 $g(x)$ 的任意相邻零点 $x_1, x_2 (x_1 < x_2)$ $g(x)$ 在 (x_1, x_2) 上单调

$\therefore [x_1, x_2]$ 上, $g(x)_{\max} = \max\{g(x_1), g(x_2)\}$

而对 $g'(x) = 0$, $f'(x) = 0$ 或 $2f'(x)f''(x) - (f'(x))^2 = 0$ 从而有 $\frac{(f'(x_0))^2}{f(x_0)} \leq 2L$ 或 $\frac{[f'(x_0)]^2}{f(x_0)} \Rightarrow f''(x_0) \leq 2L$

$\therefore$ 在 $[x_1, x_2]$ 上, $g(x) \leq \max\{g(x_1), g(x_2)\} \leq 2L$

① 若 $g'(x)$ 的零点集上下均无界, 则以 $g(x)$ 相邻零点为端点的闭区间可覆盖 $\mathbb{R}$,

$\therefore$ 即 $\forall x \in \mathbb{R}$, $g(x)$ 的相邻零点 x_1', x_2' , 使 $x \in [x_1', x_2']$, 从而 $g(x) \leq 2L$

② 若 $g'(x)$ 的零点集上有界, 下无界, 设最大零点为 x_0 .

则同①, $g(x) \leq 2L$ 在 $(-\infty, x_0]$ 上成立

在 $(x_0, +\infty)$ 上, 由导函数的介值性, $g'(x)$ 不变号, 显然若 $g'(x) < 0$, $g(x) \leq 2L$ 成立

若 $g'(x) > 0$, 则 $f'(x)$ 与 $2f'(x)f''(x) - (f'(x))^2$ 在 $(x_0, +\infty)$ 上同号

a) 若 $f'(x) > 0$ $2f'(x)f''(x) - (f'(x))^2$ $\therefore g(x) = \frac{(f'(x))^2}{f(x)} < 2f''(x)$

b) 若 $f'(x) < 0$, 则 $\lim_{x \rightarrow \infty} f(x) = 0$ (不然 $\lim_{x \rightarrow \infty} f(x) = f(x_0) + \lim_{x \rightarrow \infty} \int_{x_0}^x f'(x) dx = f(x_0) - \infty < 0$, 矛盾)

若 $\lim_{x \rightarrow \infty} f'(x) = 0$ 则 $\lim_{x \rightarrow \infty} g(x) = \lim_{x \rightarrow \infty} \frac{(f'(x))^2}{f(x)} = \lim_{x \rightarrow \infty} \frac{2f'(x)f''(x)}{f'(x)} = \lim_{x \rightarrow \infty} 2f''(x) \leq 2L$

若 $\lim_{x \rightarrow \infty} f'(x) \neq 0$, 则 $\lim_{x \rightarrow \infty} g(x) = 0 \leq 2L$

又: 在 $(x_0, +\infty)$ 上 $g'(x) > 0$, $\lim_{x \rightarrow \infty} g(x) \leq 2L$ $\therefore$ 在 $(x_0, +\infty)$ 上 $g(x) \leq 2L$

综上, $g(x)$ 在 $\mathbb{R}$ 上满足 $g(x) \leq 2L$

③ 若 $g'(x_0)$ 的零点集下有界, 同①②,

综上 $g(x) = \frac{(f'(x))^2}{f(x)} \leq 2L$

2. $\because f \in C^n([0,1])$, 且 $f^{(k)}(x) \neq 0 \quad \forall x \in (0,1)$

$\therefore f^{(k)}(x)$ 在 $[0,1]$ 上不变号

设多项式函数 $g(x)$, $\deg g(x) = n$

$$\text{令 } |g^{(n)}(x)| = M \quad g^{(n)}(x) f^{(n)}(x) \geq 0$$

$$0 \leq k \leq n-1 \quad \text{若 } f^{(k)}(x) f^{(k+1)}(x) \geq 0, \quad \text{令 } g^{(k)}(0) = f^{(k)}(0)$$

$$\text{若 } f^{(k)}(x) f^{(k+1)}(x) \leq 0, \quad \text{令 } g^{(k)}(1) = f^{(k)}(1)$$

由此可得一方程组, 进而解得 $g(x)$ 且 x^n 的系数为 $\pm \frac{M}{n!}$ (符号同 $f^{(n)}(x)$)

下面证明 对 $\forall k \in \{0, 1, 2, \dots, n\}$, $x \in [0,1]$ $f^{(k)}(x) g^{(k)}(x) \geq 0$ 且 $|g^{(k)}(x)| \leq |f^{(k)}(x)|$ ①

$$(1) k=n \text{ 时}, \because |g^{(n)}(x)| = M, g^{(n)}(x) f^{(n)}(x) \geq 0 \quad \therefore |f^{(n)}(x)| \geq M = |g^{(n)}(x)|$$

$$(2) \text{ (归纳法) 若 } k=a \text{ 时 } ① \text{ 成立, 则 } f^{(a)}(x) g^{(a)}(x) \geq 0, |g^{(a)}(x)| \leq |f^{(a)}(x)|$$

(3) 当 $k=a-1$ 时, 对 $f^{(a-1)}(x)$, $f^{(a)}(x)$ 不同的正负情况, 仅讨论 $f^{(a)}(x) \geq 0, f^{(a-1)}(x) \geq 0$ 一种, 其余情形同理

$$\therefore f^{(a)}(x) f^{(a-1)}(x) \geq 0 \quad \therefore f^{(a-1)}(0) = g^{(a-1)}(0)$$

$$\text{设 } F(x) = f^{(a-1)}(x) - g^{(a-1)}(x) (0 \leq x \leq 1) \text{ 且 } F'(x) = f^{(a)}(x) - g^{(a)}(x) \geq 0$$

$$\therefore F(x) \geq F(0) = 0 \quad \therefore f^{(a-1)}(x) \geq g^{(a-1)}(x) \quad \therefore |f^{(a-1)}(x)| \geq |g^{(a-1)}(x)|$$

综合 (1), (2), 可证明 ①

$$\text{从而 } |f(1) - f(0)| = |f(1) - f(0)| \geq |g(1) - g(0)| \quad \text{设 } h_k(x) = \frac{M}{(n-k)!} x^{n-k}$$

$$\text{再设 } h_k(x) = \frac{M}{(n-k)!} x^{n-k} \quad \text{对 } 0 \leq k \leq n-1, \text{ 若 } f^{(k)}(0) = g^{(k)}(0) \quad \text{设 } h_k(x) = \frac{M}{(n-k)!} x^{n-k}$$

$$\text{若 } f^{(k)}(1) = g^{(k)}(1) \quad \text{设 } h_k(x) = \frac{M}{(n-k)!} (1-x)^{n-k}$$

$$\text{下证 对 } k \in \{0, 1, \dots, n-1\}, x \in [0,1], |g^{(k)}(x)| \geq h_k(x) \quad ②$$

(1) $k=n$ 时, 显然成立 $k=n-1$ 时,

(2) 若 $k=a$ 时成立, 对 $k=a-1$

$$a) \text{ 若 } g^{(a)}(x) \geq 0, g^{(a-1)}(x) \geq 0 \text{ 时, 对 } G(x) = g^{(a)}(x) - h_{a-1}(x) \quad G'(x) = g^{(a+1)}(x) - h_{a-1}'(x) \geq 0$$

$$\therefore G(x) \geq G(0) = g^{(a)}(0) - h_{a-1}(0) \geq 0$$

$$b) \text{ 若 } g^{(a)}(x) \leq 0, g^{(a-1)}(x) \geq 0 \text{ 时}$$

$$a) \text{ 若 } g^{(a)}(0) = f^{(a)}(0), g^{(a-1)}(0) = f^{(a-1)}(0) \text{ 或 } g^{(a)}(1) = f^{(a)}(1), g^{(a-1)}(1) = f^{(a-1)}(1)$$

仅讨论 $g^{(a)}(0) = f^{(a)}(0), g^{(a-1)}(0) = f^{(a-1)}(0)$ 且 $g^{(a)}(x) \geq 0, g^{(a-1)}(x) \geq 0$ 一种情况, 其他情况同理

$$\text{设 } H(x) = g^{(a-1)}(x) - h_{a-1}(x) \quad \therefore H'(x) = g^{(a)}(x) - h_{a-1}'(x) \geq 0 \quad \therefore H(x) \geq H(0) \geq 0 \quad \therefore |g^{(a-1)}(x)| \geq h_{a-1}(x)$$

$$b) \text{ 若 } g^{(a)}(0) = f^{(a)}(0), g^{(a-1)}(1) = f^{(a-1)}(1) \text{ 或 } g^{(a)}(1) = f^{(a)}(1), g^{(a-1)}(0) = f^{(a-1)}(0)$$

仅讨论 $g^{(a)}(1) = f^{(a)}(1), g^{(a-1)}(0) = f^{(a-1)}(0)$ 且 $g^{(a)}(x) \geq 0, g^{(a-1)}(x) \geq 0$ 一种情况, 其他情况同理

$$\therefore g^{(a-1)}(x) = \int_0^x g^{(a)}(t) dt + g^{(a-1)}(0) \geq \int_0^x h_{a-1}'(t) dt + g^{(a-1)}(0) = h_{a-1}(x)$$

Date . . .

综上 ②成立 $\therefore |g(1) - g(0)| = \left| \int_0^1 g'(x) dx \right| \geq \left| \int_0^1 h_1(x) dx \right| \geq \frac{M}{n!}$

$\therefore |f(1) - f(0)| \geq \frac{M}{n!}$

又 $\because f(x)$ 在 $[0, 1]$ 上单调 且 $f(0), f(1)$ 同号

$\therefore |f(x)|_{\max} = \max\{f(0), f(1)\} \quad |f(x)|_{\min} = \min\{f(0), f(1)\}$

$\therefore |f(x)|_{\max} = |f(x)|_{\min} + |f(1) - f(0)| \geq 0 + \frac{M}{n!} = \frac{M}{n!}$

$\therefore \max_{x \in [0, 1]} |f(x)| \geq \frac{M}{n!}$